

RESEARCH

Writing Research That Lands: Structure & Craft for Decision- Support Reports

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CHAPTER 01

Executive Summary

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Executive Summary

Strong research reports are not distinguished by thoroughness but by **discipline of structure and narrative**: they lead with the answer, sustain one governing thought, and close with synthesis rather than restatement [HIGH]. The single most damaging structural error — in both junior-analyst and AI-generated writing — is **collapsing the Finding / Conclusion / Recommendation triad** into one undifferentiated block: *findings = what was observed, conclusions = what it means, recommendations = what to do* must be three distinct, sequential layers [HIGH]. The three most-confused sections each have a categorically distinct job: the **Executive Summary** is a standalone decision brief written last and placed first; the **Introduction** is the motivational gateway (why the research, what it seeks to learn, where it stops); the **Conclusion** is the synthesized "so what" at the end — never a repeat of the summary [HIGH]. The recommended template for IW is **decision-first, rigor-backed**: a BLUF/Pyramid-Principle answer up front, IMRaD-grade methodology and confidence transparency underneath, and every finding tagged HIGH/MEDIUM/LOW [HIGH]. **For this project specifically, the fix is to add a structured Introduction, an expanded Background, an Analysis/Discussion step, and a mandatory Conclusion to the /iw- research skill, and to instruct the agent on prose craft (declarative headings, BLUF, synthesis, calibrated confidence), because LLMs default to an academic build-toward-the-conclusion structure unless explicitly overridden** [MEDIUM].

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CHAPTER 02

Introduction

CHAPTER 02

Introduction

The IW research module recently gained visualizations ([R-00153](#)), which measurably improved its output. But review of the resulting documents (notably [R-00157](#)) exposed a deeper weakness: the **prose scaffolding is thin**. The Background was a single cross-reference line, there was **no Conclusion at all**, and the documents read as findings dumps rather than reasoned arguments. Diagrams made the research prettier; they did not make it *think* better. This research was commissioned to fix that — to define exactly how a high-quality research report should be structured and written, so the skill can be rebuilt to that standard.

Why this research

Objectives — what we expect to learn

1. The **canonical section taxonomy** of a research report and the distinct role each section plays.
2. Crisp, authoritative definitions distinguishing the three most-confused sections — **Executive Summary, Introduction, Conclusion** — and what makes a strong Conclusion specifically (the current gap).
3. The **writing-craft** rules (narrative thread, clarity, calibrated confidence, actionable recommendations) that separate a report that lands from one that is merely complete.
4. A concrete, **agent-followable specification** for the new `/iw-research` output structure in a "hybrid: analyst-led, rigor-backed" style.

Scope

In scope: report structure, section roles, and prose craft for decision-support research. Out of scope: data visualization and chart selection (covered by [R-00153](#)) and the diagram tool landscape ([R-00051](#)). This document deliberately **uses the structure it recommends**, as a working reference example.

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CHAPTER 03

Background & Context

CHAPTER 03

Background & Context

A research report is a *genre* with conventions drawn from three traditions, each contributing a different strength:

- **Scientific writing** contributes **IMRaD** (Introduction, Methods, Results, and Discussion), formalized by ANSI in 1972 and near-universal in biomedical journals — its value is rigor and navigability ([UBC Science Writing](#), [AMWA](#)).
- **Consulting/analyst writing** contributes the **Minto Pyramid Principle** (McKinsey, 1960s) and **BLUF** (bottom line up front, from US military intelligence) — its value is decision-speed: the answer comes first ([Think Insights](#), [Animalz on BLUF](#)).
- **Intelligence analysis** contributes the **confidence discipline** — a three-tier HIGH/MODERATE/LOW system that separates evidence from judgment ([CIS MS-ISAC](#)).

The "hybrid" target combines all three: lead with the answer (consulting), prove it transparently (scientific), and calibrate every claim (intelligence).

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CHAPTER 04

Methodology

CHAPTER 04

Methodology

Two parallel deep-research passes were run — one on report **structure/anatomy**, one on writing **craft/narrative** — across university writing centers (USC, Purdue OWL, Harvard, GMU, Indiana, UNC), professional authorities (Gopen & Swan, Pinker, Minto, NN/G, Plain Language guidelines), and analyst/intelligence standards (CIS/ICD 203). Findings are tagged with a three-tier confidence scale: **HIGH** = multiple authoritative corroborating sources; **MEDIUM** = credible but limited corroboration; **LOW** = single-source or inference. This scale is the one recommended for the tool itself (F6).

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CHAPTER 05

Findings

CHAPTER 05

Findings

Figure 1. The canonical report anatomy in three zones — orientation, framing, substance. Cross-referenced across Purdue OWL, USC, U. Reading, and ANU report guides ([Purdue OWL](#), [U. Reading](#)).

F1 — A research report has a canonical section anatomy in three zones [HIGH]

Why: the section set is the skeleton everything else hangs on; naming the zones clarifies what each section is *for*.

Shorter reports embed Objectives/Scope inside the Introduction and may merge Analysis into Findings; longer decision-support documents promote them to standalone sections ([ANU](#)).

F2 — The Finding / Conclusion / Recommendation triad must never collapse [HIGH]

The crispest formulation in the literature: **Findings = what was observed; Conclusions = what it means; Recommendations = what to do about it** ([EvalCommunity](#)). A finding is a factual statement from evidence; a conclusion is the interpretive judgment drawn from it; a recommendation is the action it justifies ([Journalism University](#)). Collapsing these three into one "findings" block is the single most common structural error and **eliminates the report's analytical value** — the reader can no longer tell observation from inference from prescription. Recommendations follow the formula "*To help with X, we recommend Y, because of Z*" ([SlideModel](#)).

F3 — Executive Summary, Introduction, and Conclusion do three different jobs [HIGH]

Why: these three are constantly conflated; the table below is the reference an agent should follow to keep them distinct.

Dimension	Executive Summary	Introduction	Conclusion
Answers	"What should I know & do?"	"Why this, and what will it learn?"	"What does the evidence mean?"
Position	First	Second	Near last
Written	Last	Early	After Findings/Analysis
Contains conclusions?	Top-line only	No	Yes — full synthesis
Contains methodology?	One sentence	Brief	No
Standalone?	Yes — by design	No	No
Most common error	Written as a table of contents	States conclusions instead of questions	Restates findings instead of synthesizing
Length	5–10%	5–10%	3–5%

The Executive Summary must let a reader **act without reading the body** — it is "not an introduction" ([USC Exec Summary](#)). The Introduction establishes *why the research, what it seeks to learn, and where it stops*, and must not use conclusion-register words like "found" or "demonstrates" ([GMU IMRaD](#)).

F4 — A strong Conclusion synthesizes and answers "so what" — it never restates [HIGH]

This is the project's key gap. The Conclusion "synthesizes key points from your analysis rather than merely summarizing topics" and its defining obligation is to answer **"So What?"** ([USC](#)

[Conclusion](#)). Harvard frames it as three moves — **what / so what / now what** ([Harvard Writing Center](#)). A strong conclusion: restates the research question, states the answer the findings compel, draws implications beyond any single finding, and introduces **no new evidence**. The inverse of the Introduction — where the Introduction moves general → specific, the Conclusion moves specific findings → broader significance. Forbidden: new data, section-by-section recaps, clichéd "In conclusion...", and apologetic hedging (that belongs in Limitations).

F5 — Background ≠ Introduction; Scope ≠ Limitations [HIGH]

Background supplies domain context and prior work needed to understand the problem; the **Introduction** sets up the research itself (problem, questions, scope) ([USC Background](#)). **Scope/Delimitations** are deliberate boundary choices (belong in the Introduction); **Limitations** are constraints outside the author's control that qualify the findings (belong near the end) ([AJE on scope vs delimitations](#)). Conflating either pair prevents readers from separating deliberate focus from accidental gaps.

F6 — Methodology is the transparency anchor; confidence labels make trust calibratable [HIGH]

Methodology documents how the evidence was gathered (primary vs secondary, sources, analysis) and **declares the confidence framework** so readers can interpret the HIGH/MEDIUM/LOW tags used throughout ([CIS analytic confidence](#)). Every claim should be assignable to one of three categories — **evidence** (a cited datum), **inference** ("the evidence suggests..."), or **assumption** (explicitly flagged) — and conflating them is the most corrosive form of analytical dishonesty. Transparency about method and limitations is a **credibility amplifier, not a confession of weakness** ([Gopen & Swan](#)).

F7 — Lead with the answer: the Pyramid Principle, SCQA, and BLUF [HIGH]

The **governing thought** — the single most important idea, the answer to the reader's primary question — must appear first; supporting points are *summaries of the evidence beneath them*, ideally grouped **MECE** (mutually exclusive, collectively exhaustive) ([Think Insights](#)). The **SCQA** sequence (Situation → Complication → Question → Answer) structures the opening: it maps Situation → Background, Complication → problem statement, Question → objectives, Answer → Executive Summary ([ModelThinkers](#)). This is the deductive inverse of academic inductive structure — and the default LLMs must be told to override (F10).

F8 — Prose clarity is a structural discipline, not a flourish [HIGH]

Gopen & Swan's reader-expectation theory: readers weight information at the **stress position** (sentence end) and orient from the **topic position** (sentence start) using known information — so put the familiar first and the new/important last, and keep subject and verb close ([Science of Scientific Writing](#)). Plain-language standards: average sentence < 20 words, active voice, one idea per paragraph with the topic sentence first ([plainlanguage.gov](#)). Pinker names the

curse of knowledge — the writer's inability to imagine not knowing what they know — as the chief cause of bad expert prose; the antidote is concrete "classic style" and defining every term on first use ([Pinker](#)).

F9 — Declarative headings and scannability carry disproportionate weight [MEDIUM]

Readers scan digital documents in F/Z patterns, front-loading headings and opening sentences; concise, scannable, objective writing measured **124% more usable** than promotional prose in NN/G's study ([NN/G](#)). Implication: headings should be **declarative findings** ("Vendor X leads on latency but trails on cost"), not topic labels ("Vendor comparison"). The headings, read alone in sequence, should reconstruct the report's argument. Pair data with implication inline using the **data → finding → insight → recommendation** hierarchy ([NN/G data vs insights](#)).

F10 — LLMs default to academic structure and need explicit, per-section craft instruction [MEDIUM]

LLMs over-trained on academic papers default to inductive *context → evidence → conclusion* rather than decision-first structure, and under-invest in the conclusion; long-form-report evaluations flag narrative coherence, synthesis quality, and citation accuracy as the weakest dimensions ([Deep-Research Eval, MDPI](#)). The fixes are procedural: (a) **separate generation passes** — gather → draft-with-governing-thought → editorial-craft review; (b) **per-section instructions** rather than one "write a good report"; (c) require a governing thought before drafting; (d) enforce BLUF and declarative headings; (e) give the Conclusion a dedicated word budget and a "so what" check.

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CHAPTER 06

Analysis / Discussion

CHAPTER 06

Analysis / Discussion

Three threads converge. **First**, every framework above is the *same rule at different scales*: BLUF at the document level, the governing thought at the section level, and the stress/topic positions at the sentence level all say *put the point where the reader looks for it*. **Second**, the failure modes are predictable and therefore checklistable — findings dumps, buried ledes, weak conclusions, vague recommendations, and uncalibrated confidence recur across genres and are each correctable by a specific rule. **Third**, and most relevant to this project: the gaps observed in the EU AI Act suite were not random — they are exactly the gaps the literature predicts for AI-generated reports (thin framing, absent synthesis, academic default order). That means they are fixable structurally, by changing the skill's mandated template and per-section prompts, rather than hoping for better free-form generation.

A genuine tension surfaced: rigor (methodology, hedging, limitations) versus decision-speed (BLUF, assertive recommendations). The resolution in the hybrid model is *sequencing, not compromise* — assert the answer up front for skimmers, then expose the rigor beneath it for scrutinizers, and calibrate with confidence tags so honesty does not read as waffling.

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CHAPTER 07

Conclusion

CHAPTER 07

Conclusion

A research report lands when its structure mirrors how decision-makers actually read — **answer first, evidence beneath, synthesis at the end** — and when it never blurs the line between what was observed, what it means, and what to do. The evidence compels a specific answer to this project's question: the IW research module's weakness is **structural, not cosmetic**, and is best fixed by mandating four additions to the skill — a framed **Introduction** (why/objectives/scope), an expanded **Background**, an **Analysis/Discussion** step, and a mandatory **Conclusion** that synthesizes and answers "so what" — together with **per-section prose-craft instructions** (declarative headings, BLUF, calibrated confidence, the finding/conclusion/recommendation firewall). The broader implication is that quality in AI-generated research is achievable not by asking the model to "write well," but by encoding the genre's hard-won conventions as explicit, checkable structure. The most important single change is the one the current output lacks entirely: **a conclusion that thinks, rather than repeats**. The chief remaining uncertainty is calibration — how heavy the framing should be for short versus long research — which the skill should handle by scaling sections to depth rather than mandating every section at every length.



CHAPTER 08

Recommendations

CHAPTER 08

Recommendations

1. **[Structure] Adopt the three-zone canonical template** (Figure 1) as the mandated */iw-research* output, scaling sections to depth (short research may merge Analysis into Findings and embed Scope in the Introduction). *Supports F1.*
2. **[Conclusion — highest priority] Make a synthesizing Conclusion mandatory.** Its first sentence must restate the primary question as a declarative answer; it must state implications ("so what") and introduce no new evidence. *Supports F4, the core gap.*
3. **[Introduction] Require an Introduction with explicit *Why / Objectives / Scope*** and forbid conclusion-register words ("found", "shows") in it. *Supports F3, F5.*
4. **[Firewall] Keep Findings, Conclusion, and Recommendations as distinct labeled sections;** every recommendation must trace to a finding/conclusion ("To help with X, we recommend Y because Z"). *Supports F2.*
5. **[Confidence] Keep per-finding HIGH/MEDIUM/LOW tags and declare the scale in Methodology;** separate evidence from inference from assumption. *Supports F6.*
6. **[Craft] Add per-section prose rules to the skill:** BLUF openings, declarative finding headings, active voice, define terms on first use, weave citations with signal phrases. *Supports F7–F9.*
7. **[Process] Instruct the agent to write the Executive Summary last** and to apply an editorial "so what / does the conclusion synthesize?" self-check before finishing. *Supports F10.*
8. **[Retrofit] Regenerate R-00153–R-00157 to the new template** so the live examples model the standard.

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CHAPTER 09

Limitations

CHAPTER 09

Limitations

- Guidance on **AI-specific** report quality is an emerging area (2025–2026 sources); the per-section prompting recommendations are well-grounded in general methodology but not yet standardized for AI pipelines.
 - The Minto Pyramid Principle and BLUF are practitioner frameworks demonstrated by adoption rather than controlled comprehension studies.
 - NN/G scannability findings derive from web content; long-form PDF reading may differ, though front-loading holds across media.
 - Confidence vocabulary is adapted from intelligence/cyber analysis (ICD 203/CIS) and may need calibration for non-analyst domains.
 - This research covers prose and structure only; visual design is covered by [R-00153](#).
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CHAPTER 10

Sources

CHAPTER 10

Sources

#	Title	Credibility	URL
1	Purdue OWL — Sections of Reports	High (academic authority)	https://owl.purdue.edu/owl/subject_specific_writing/writing_in_engineering/handbook_on_report_format/
2	USC — Executive Summary	High (university guide)	https://libguides.usc.edu/writingguide/executivesummary
3	USC — Background Information	High (university guide)	https://libguides.usc.edu/writingguide/background
4	USC — The Conclusion	High (university guide)	https://libguides.usc.edu/writingguide/conclusion
5	University of Reading — Structuring your report	High (university guide)	https://libguides.reading.ac.uk/reports/structuring
6	ANU — Report structure	High (university guide)	https://www.anu.edu.au/students/academic-skills/writing-assessment/report-writing/structure
7	GMU Writing Center — Writing an IMRaD Report	High (university guide)	https://writingcenter.gmu.edu/writing-resources/imrad/writing-an-imrad-report
8	Søk og Skriv — The IMRaD format	High (academic consortium)	https://www.sokogskriv.no/en/writing/the-imrad-format.html
9	UBC — IMRaD science writing	High (university guide)	https://scwrl.ubc.ca/stem-writing-resources/features-of-academic-stem-writing/imrad/
10	AMWA — IMRaD format explained	High (medical writers assoc.)	https://blog.amwa.org/imrad-format-explained
11	EvalCommunity — Finding vs Conclusion vs Recommendation	Medium-High	https://www.evalcommunity.com/career-center/differences-between-a-finding-a-conclusion-and-a-recommendation/
12	Journalism University — Conclusions & Recommendations	Medium	https://journalism.university/communication-research-methods/research-report-summary-conclusions/
13	Harvard College Writing Center — Conclusions	High (university guide)	https://writingcenter.fas.harvard.edu/conclusions
14	Think Insights — Pyramid Principle	Medium (consulting synthesis)	https://thinkinsights.net/strategy/pyramid-principle

15	ModelThinkers — Minto Pyramid & SCQA	Medium	https://modelthinkers.com/mental-model/minto-pyramid-scqa
16	Animalz — BLUF (Bottom Line Up Front)	Medium	https://www.animalz.co/blog/bottom-line-up-front
17	SlideModel — Consulting report guide	Medium	https://slidemodel.com/consulting-report-how-to-write-and-present-one/
18	CIS MS-ISAC — Words of Estimative Probability & Analytic Confidence	High (gov cybersecurity)	https://www.cisecurity.org/ms-isac/services/words-of-estimative-probability-analytic-confidences-and-structured-analytic-techniques
19	AJE — Scope and Delimitations in Research	High (scientific editing)	https://www.aje.com/arc/scope-and-delimitations-in-research
20	Gopen & Swan — The Science of Scientific Writing (UCL PDF)	High (peer-reviewed, 1990)	https://www.gatsby.ucl.ac.uk/~pel/misc/gopen_swan.pdf
21	plainlanguage.gov — Federal Plain Language Guidelines	High (US Government)	https://www.plainlanguage.gov/howto/guidelines/FederalPLGuidelines/FederalPLGuidelines.pdf
22	Pinker — The Curse of Knowledge (APS Observer)	High (peer journal)	https://www.psychologicalscience.org/observer/the-curse-of-knowledge-pinker-describes-a-key-cause-of-bad-writing
23	NN/G — Concise, Scannable, Objective	High (UX research authority)	https://www.nngroup.com/articles/concise-scannable-and-objective-how-to-write-for-the-web/
24	NN/G — Data, Findings, and Insights	High (UX research authority)	https://www.nngroup.com/articles/data-findings-insights-differences/
25	NN/G — Text Scanning Patterns (eyetracking)	High (UX research authority)	https://www.nngroup.com/articles/text-scanning-patterns-eyetracking/
26	Indiana University — Paragraphs and Topic Sentences	High (university guide)	https://wts.indiana.edu/writing-guides/paragraphs-and-topic-sentences.html
27	UNC Writing Center — Flow	High (university guide)	https://writingcenter.unc.edu/tips-and-tools/flow/
28	Drillbit Labs — Effective		

	recommendations (FIRE)	Medium	https://depth.drillbitlabs.com/p/research-recommendations
29	Ohio State — Executive Summary and Abstract	High (university OER)	https://ohiostate.pressbooks.pub/feptechcomm/chapter/5-2-executive-summary-abstract/
30	Deep-Research Eval — long-form report quality (MDPI)	Medium (peer-reviewed, 2025)	https://www.mdpi.com/2076-3417/16/5/2546